



UNIVERSITY OF WEST ATTICA
DEPARTMENT OF INFORMATICS AND COMPUTER ENGINEERING
MSc PROGRAM *ADVANCED COMPUTING SYSTEMS*

MSc topics 2024-2025
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1. Performance evaluation of vector databases

Recommendation systems are crucial in a plethora of sectors including e-commerce, digital culture, document processing systems, and social media. At the heart of these systems lie efficient knowledge representation and data point similarity schemes. Vector databases are an emerging technology where geometry plays an important role in determining clusters of data points with high contextual coherence. Within the bounds of the proposed MSc thesis the applications of vector databases to recommendation systems will be studied with emphasis placed on performance metrics in similarity searches including precision and accuracy. Specifically, the three primary research objectives are to compile the recent scientific literature about the field, to provide a comprehensive comparison of four mainstay databases, namely Faiss, Chroma, Pinecone, and Milvus, including their applications to machine learning. and to develop a recommendation system for a real world collection as a concrete example. The importance of the proposed topic stems from the facts that it is a field still under formulation, has tremendous potential across applications, and the application will be a point of reference for practitioners and researchers alike.